

Modern Business Cycle Theory

Measurement, expectations and a consumption–saving model

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SESSION OBJECTIVE

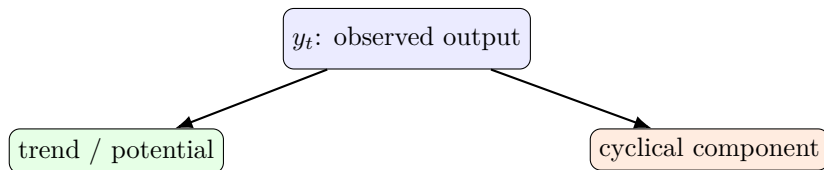
- ▶ Move from observed macroeconomic series to a disciplined notion of a business cycle.
- ▶ Use a simple intertemporal model to connect shocks, expectations and aggregate fluctuations.
- ▶ Prepare the approximation tools used throughout the course.

Handout: Session 1 **Code:** `code/session1/`

PLAN

- 1 What is a cycle?
- 2 A dynamic benchmark
- 3 From nonlinear model to usable system
- 4 Takeaways

OBSERVED OUTPUT COMBINES SEVERAL OBJECTS



A filter is an *operational definition*: its assumptions determine what counts as trend, cycle and measurement noise.

THREE QUESTIONS BEFORE FILTERING

1. What frequency range corresponds to the economic question?
2. Is the trend deterministic, stochastic or model based?
3. What happens at the sample endpoints and after revisions?

Course principle

Filtering is not neutral preprocessing. It is part of the empirical model and must be documented.

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A CONSUMPTION–SAVING PROBLEM

Household choice:

$$\max_{\{c_t, a_{t+1}\}} \mathbb{E}_0 \sum_{t \geq 0} \beta^t u(c_t)$$

subject to

$$c_t + a_{t+1} = y_t + (1 + r_t)a_t.$$

- ▶ β : patience
- ▶ a_t : assets carried into t
- ▶ r_t : intertemporal price
- ▶ y_t : stochastic income

Economic force: households smooth consumption across states and dates.

EULER EQUATION: THE KEY RESTRICTION

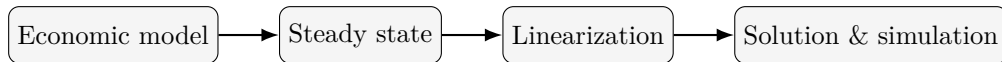
$$u'(c_t) = \beta \mathbb{E}_t [(1 + r_{t+1})u'(c_{t+1})].$$

- ▶ Current consumption trades off against expected marginal utility tomorrow.
- ▶ Expectations are endogenous: beliefs concern future equilibrium objects.
- ▶ Estimation later asks which parameters make these restrictions fit the data.

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A REPRODUCIBLE WORKFLOW



- ▶ Keep the nonlinear model and the approximation separate.
- ▶ Check residuals and units before interpreting an impulse response.

FIRST-ORDER APPROXIMATION

For a system $F(x_{t+1}, x_t, x_{t-1}, \varepsilon_t) = 0$, expand around the deterministic steady state:

$$F_x \hat{x}_{t+1} + F_y \hat{x}_t + F_z \hat{x}_{t-1} + F_\varepsilon \varepsilon_t \simeq 0.$$

Why it matters

Linear systems make expectations, filtering and likelihood evaluation computationally tractable.

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TAKEAWAYS AND PREPARATION

- ▶ Business-cycle facts depend on measurement choices.
- ▶ Dynamic macroeconomic models express restrictions through Euler equations and expectations.
- ▶ Linearization supplies the state-space objects used in Sessions 2–4.

Before Session 2: run `toy_model.m` and inspect the model, steady state and simulated paths.